

Final Report Team 3 Retention Radar

Retention Radar - Customer Churn Analysis for Advanced Telecommunications Solutions (ATS)

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Purpose: This document directly addresses the Final_Report checklist items: key findings, churn and retention factors, targeted retention recommendations, and limitations with proposed solutions.

Executive Summary

The Stage 3 ANN model provides strong business value because it converts customer data into risk ranking, churn-driver insight, and actionable retention segments. The main value is not only the model score, but the ability to identify which customer groups should be prioritised for retention activity.

- The refined ANN model achieved 73.39% accuracy, 73.46% recall, 59.44% F1-score, 0.8172 ROC-AUC, and 0.5922 PR-AUC at the selected decision threshold of 0.52.
- The strongest churn drivers are MonthlyCharges and tenure, with EstimatedTotalCharges also contributing but at a much lower level.
- The riskiest 20% of customers contain 44.7% of all actual churners, supporting targeted retention rather than a blanket campaign.
- The short-tenure, high-charge customer segment has the highest actual churn rate at 48.9%, making it a priority group for retention action.
- Data Engineer QA confirms the Stage 3 hand-off data is reliable: all checks passed, with no missing values, no duplicates, no train/test overlap, and correct scaling.

A. Summary of Key Findings

This section summarises the key findings from customer segmentation and churn prediction analysis conducted using the ANN model.

A1. ANN Churn Prediction Results

The final ANN model was tested on 1,349 unseen customers. The model is useful for business decision-making because it identifies a large proportion of actual churners and ranks customers by churn risk.

Metric	Final result	Business meaning
Accuracy	73.39%	The model correctly classifies a strong share of overall customer outcomes.
Precision	49.91%	Around half of customers flagged as churners were actual churners, so contact strategy should remain targeted.
Recall	73.46%	The model catches about three out of four actual churners, making it useful for retention planning.
F1-score	59.44%	Shows a reasonable balance between precision and recall.
ROC-AUC	0.8172	Strong ability to rank risky customers above safer customers.
PR-AUC	0.5922	Useful for an imbalanced churn problem where churners are the minority.
Decision threshold	0.52	Selected cut-off used to balance churn detection and contact volume.

Figure 1. Model performance summary used for final report interpretation.

A2. Customer Segmentation Findings

The segmentation analysis connects churn predictions to practical customer groups. The highest-risk group is short-tenure, high-charge customers, who are commercially important because they have high monthly value but have not yet developed long-term loyalty.

Customer segment	Customers	Average tenure	Average monthly bill	Actual churn
Short tenure, high charges	419	15.2 months	\$81.98	48.9%
Short tenure, low charges	336	11.0 months	\$33.10	23.5%
Long tenure, high charges	390	58.4 months	\$91.98	16.7%
Long tenure, low charges	204	54.8 months	\$33.63	4.4%

Figure 2. Churn rate by customer segment.

A3. Risk Ranking and Risk Bands

The model ranked customers by predicted churn risk. The most useful business insight is that the riskiest 20% of customers contain 44.7% of actual churners. This means the business can focus retention activity where it is most likely to matter.

Risk band	Customers	Share of customer base	Actual churn rate	Share of all leavers
High risk above 0.70	284	21.1%	58.8%	46.6%
Medium risk 0.40-0.70	404	29.9%	34.9%	39.4%
Low risk below 0.40	661	49.0%	7.6%	14.0%

Figure 3. Risk-band action list for retention planning.

B. Identification of Factors Contributing to Churn and Retention

This section identifies and analyses the main factors contributing to customer churn and retention based on the ANN model predictions.

Factor	Model finding	BA analysis / business interpretation
MonthlyCharges	Strongest churn signal with importance 0.138.	Higher monthly charges may create price pressure or reduce perceived value. Customers with high bills should be reviewed for plan fit, billing clarity, and retention support.
Tenure	Second strongest signal with importance 0.119.	Newer customers have higher churn risk because loyalty and relationship value have not yet developed. First-year engagement is critical.
EstimatedTotalCharges	Third driver with importance 0.013.	This helps represent customer value and relationship history, but it is derived from tenure and monthly charges rather than actual billing history.
SeniorCitizen	Minor signal with importance 0.011.	May help with risk context but should not be used alone for targeting.
IsNewCustomer	Minor signal with importance 0.009.	Supports the finding that early-tenure customers need proactive onboarding and support.
Contract type	Approximately zero model importance in this dataset.	This is unusual for telecom churn, so recommendations should not rely mainly on contract type without further validation.

Figure 4. Feature importance: MonthlyCharges and tenure dominate churn prediction.

Time as customer	Customers	Actual churn rate	final report interpretation
0-12 months	417	45.8%	Highest churn risk; prioritise onboarding and early engagement.
13-24 months	176	33.0%	Still high risk; maintain proactive follow-up.
25-48 months	317	19.6%	Moderate churn risk; use value reinforcement.
49-72 months	439	10.7%	Lowest churn risk; focus on loyalty maintenance.

Figure 5. Churn decreases sharply as customer tenure increases.

C. Recommendations for Targeted Retention Strategies

This section outlines targeted retention strategies based on the ANN model results, customer segmentation, churn drivers, and risk-band analysis.

Recommendation	Target group	Action	Business reason
Prioritise the top 20% highest-risk customers	Customers in the highest risk deciles	Create a priority retention contact list using predicted churn risk scores.	This group contains 44.7% of actual churners, so it gives the best return for retention effort.
Focus on the first 12 months	Customers with 0-12 months tenure	Use onboarding calls, early check-ins, service education, and welcome offers.	First-year customers show 45.8% churn, making early engagement essential.
Review high-charge, short-tenure customers	Short-tenure, high-charge segment	Offer plan-fit reviews, bill explanation, value checks, and personalised support.	This segment has 48.9% actual churn and represents high-value revenue at risk.
Use risk-band action planning	High, medium, and low risk bands	Use stronger offers for high-risk customers and lower-cost digital engagement for medium-risk customers.	Matches retention cost to customer risk and avoids overspending on low-risk customers.
Do not rely only on contract type	All customer groups	Base targeting on risk score, tenure, and monthly charges rather than contract type alone.	Contract type did not show meaningful signal in this dataset.
Use model as decision support	Retention and business stakeholders	Combine ANN risk scores with customer service history, complaint context, and commercial judgement.	Prevents over-reliance on a black-box model and improves fairness of retention decisions.

D. Documentation of Limitations and Proposed Solutions

This section documents limitations, challenges, and proposed solutions identified during Stage 3. These should be stated clearly so stakeholders understand how to use the model responsibly.

Limitation / challenge	Impact	Proposed solution
Missing customer service fields	The dataset does not include support tickets, complaint history, customer satisfaction, payment method, competitor activity, or actual billed total charges.	Collect richer customer service, billing, complaint, competitor, and satisfaction data in future iterations.
EstimatedTotalCharges is derived	It is calculated from tenure and monthly charges, not from actual billing history.	Use actual billing and payment history where available.
ANN explainability	Neural networks can be harder for non-technical stakeholders to interpret.	Use feature importance, segment analysis, risk bands, and plain-English explanations in reports and presentations.
False positives and false negatives	Some loyal customers may be contacted unnecessarily, and some churners may still be missed.	Select decision threshold based on retention budget, contact capacity, and tolerance for missed churners.

Contract type result is unusual	Contract type did not strongly predict churn in this dataset, which differs from typical telecom expectations.	Validate with a larger or more complete dataset before using contract-focused campaigns.
Decision-support only	The model should not be the only basis for customer treatment decisions.	Use the ANN as a decision-support tool alongside human judgement, customer context, and service history.

Supporting Data Engineer QA Summary

The Data Engineer QA report is included as supporting evidence because it confirms the model results are based on a reliable Stage 3 data hand-off.

QA check	Result	BA meaning
Row counts	6,741 total; 5,392 training; 1,349 testing	The data split matches the expected project design.
Churn-rate balance	26.57% full; 26.58% training; 26.54% testing	Training and testing sets are representative.
Train/test overlap	0 overlapping rows	The model was tested on unseen customers.
Scaling	Correct; fit on training data only	Reduces risk of test-data leakage.
Missing values	0 blank values	No missing-data issue affecting the model hand-off.
Duplicate records	0 duplicates	No duplicate customer records inflating patterns.
Column checks	All feature groups present and correctly named	The model used the expected Stage 2 features.

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